

ActInf Livestream #031.0 ~ "Non-equilibrium Thermodynamics and the Free Energy Principle in Biology"

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[Laughter] hello everyone welcome to the active inference lab this is active inference live stream number 31.0 it's october 14 2021. welcome to the act imp lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams at this short link you can see some of the past present and future streams that we'll be doing in the lab as part of the dot com's organizational unit and the first tab has the regular tuesday discussions that are group discussions dot one and dot two right now we're in a dot zero kind of an ad hoc scheduled contextualizing live stream and then the other tabs have other kinds of live streams that can be scheduled at different times different topics so just let us know if you want to get involved or co-organize an event with us today in active stream number 31.0 the goal is to learn and discuss this paper non-equilibrium thermodynamics and the free energy principle in biology by patricia palacios and matteo colombo like all the dot zeros you'll get the disclaimer that this is just an introduction to some of the ideas of the paper and the context it's not a review or a judgment or a final word in 31.0 we're going to start by looking over some of the key aims and claims and the abstract and the roadmap of the paper to get the big picture then talk about some of the keywords maybe you're really familiar with some of the keywords maybe it's the first time you've thought about some of these topics and then we'll walk through some of the figures and formalisms and just get set up in the coming weeks to be discussing this paper so we'll start with just some introductions and warm-ups we'll say hello and then maybe just mention something that we're excited to talk about or learn about or something that we noticed while reading the paper so i'm daniel i'm a researcher in california and this is just an exciting paper and topic because it's at the intersection of chemistry and physics and

thermodynamics but also active inference and it's a nice starting point for a conversation about what are the precursors and the tenants and the principles of free energy principle and active inference and then what are the consequences and what does that mean for the big picture blue i'm blue knight i'm an independent research consultant in new mexico uh and i really was excited to see this paper bring up some of the points that i've brought up as i've been reading more about active inference specifically with uh respect to ergodicity and what that means the potential implications for that and also just other minor topics but things i've mentioned in the past it's cool sometimes to see how the threads of our previous discussions and things that a participant might have brought up like hey this isn't making sense and it's like don't worry it's not just you these are open research questions and so we're seeing that and we'll be enacting it in the coming weeks some of the big questions in the paper kind of framed in non-technical or in not the author's words it's really asking what fundamental ideas from physics are utilized in the free energy principle and how are they utilized why are they utilized and then what are some implications or trade-offs associated with the use of these ideas either the ways that they have been used what are the pros and cons or what are the implications and what about the ways that they could be used so in the spirit of positive contributions to the literature how are we taking ideas from physics and other fields bringing it together in free energy principle and then what are the fundamental or just spuriously contemporary tradeoffs let's first go to the aims and claims of the authors so all read it and then blue you can give a thought well the paper is the non-equilibrium thermodynamics and free energy principle in biology on the fillsci archive at pitt so check out the paper and read it if you want to and a few of their aims and claims that we can extract out the free energy principle has received a lot of attention but its foundations in statistical physics and dynamical systems theory have not been probed yet this lack of attention is unfortunate because the theoretical adequacy of the free energy principle as well as its practical utility for the study of key biological properties such as homeostasis and robustness depend on the validity of those foundations here we begin to fill this gap so that's kind of one of their contextualizing claims and their aim and one of their arguments a claim about their arguments is that as we are going to argue the foundations and physics concepts allow models built from the free energy principle to achieve maximum generality but these assumptions also decrease the ability of these models to include enough factors to provide biologically plausible representations of the causal networks responsible for living systems dynamic equilibrium any thoughts on the aims or we can continue wait i can't hear you i think that the aims are good sorry i just realized i was muted um i think that the

these are good aims i think that uh one of the fundamental um claims is that in order to make the fep more general generalizable it's not as specific to biology as it could be i think that that was like the key trade-off maybe yes between specificity and utility and generalizability so one mindset is that as we generalize we're going to be explaining more systems different systems explaining them better doing more and it's just all up up with generalization another contrasting opinion is that as we generalize we out of necessity abstract or average out over particulars and so we might actually lose utility or potentially and even disastrously lose the specifics of the system that we're talking about so all of a sudden we have this free floating generalization or abstraction that maybe initially was inspired by wanting to explain some sort of key biological property like homeostasis and robustness in the author's view but then we end up sort of throwing out the kernel or the essence of what we wanted to explain because we generalized too much so we'll be exploring that okay do you want to read the abstract sure according to the free energy principle life is an inevitable and immersion property of any ergodic random dynamical system at non-equilibrium steady state that possesses a markov blanket first in 2013. formulating a principle for the life sciences in terms of concepts from statistical physics such as random dynamical system non-equilibrium steady state and ergodicity places substantial constraints on the theoretical and empirical study of biological systems thus far however the physics foundations of the free energy principle have received hardly any attention here we start to fill this gap and analyze some of the challenges raised by applications of statistical physics from modeling biological targets based on our analysis we conclude that model building grounded in the free energy principle exacerbates a trade-off between generality and realism because of a fundamental mismatch between its physics assumptions and the properties of actual biological targets cool and there's a lot to say that's just the opening salvo from the authors how do they make good on this abstract we can look at the roadmap there's not too many sections in this paper but there's a lot of uh meat within there's a general introduction and then a specific introduction to the free energy principle which is described in the context of self-organizing dissipative systems and then there's a little discussion on the explanatory scope of the fep and the introduction of the winged snowflake thought experiment or idea which we'll return to soon section 3 talks about biological systems as random dynamical systems which isn't something that fep invented it's not something that active invented this is a broader question about how to model biological systems and dynamical systems section 4 models biological active states as random dynamical attractors and talks about three big physics ideas that if we take section three you know if we want to be

modeling biological systems as dynamical systems then we're using this physics ontology and what other key terms are relevant in physics for discussing random dynamical systems and we will go through them they are the idea of phase spaces in biology ergodicity and random dynamical attractors there's then a discussion and a short conclusion so that's kind of the overall roadmap that's where we're going to go we're going to be exploring that trade-off or the implications of the introduction of physics fundamentals into the generalization of biological systems and maybe even non-living systems as well and just how's it all going to come together and how does it influence how we research going forward the key words are fep free energy principle dynamic equilibrium homeostasis phase base ergodicity and attractor and we can ask where's active inference not that it has to be a keyword for everything but as we discuss let's keep in mind what are the differences between fep and active and what are the implications of these claims about fep for active all right how about the first keyword so interestingly like you um said that you like how this paper brings together physics and chemistry and biology and i thought that chemistry was kind of blatantly omitted and in my heart i'm a chemist like if i hadn't studied neuro i would have really like gone into ochem i just thought that like working in the lab was a little scary like too much ether and two like whoa when i came out um so so really uh when i look up dynamic equilibrium or the context that i'm familiar with dynamic equilibrium is in chemistry and so a dynamic equilibrium occurs when you have a reversible chemical reaction so it's not that the reaction is over right like here's an example of water um ionizing into hydroxide and um protons right and so you you it's not that when you have the equilibrium that the reaction is still occurring it's just occurring at equal rates in the forward and reverse directions so this is dynamic equilibrium and it only happens in a reversible chemical reaction like so so there's no other way that i know of that dynamic equilibrium um happens it is an example of a system particularly a chemical system at a steady state and so i looked through the paper i don't know if you want to flip to the next slide um i looked to the paper and i i looked like well how are the authors using and i found this um phrase which i'll say here the core of this account is a free energy principle um according to which all biological systems actively maintain a dynamic equilibrium with their environment by minimizing their free energy which enables them to avoid a rapid decay into an inert state of thermodynamic equilibrium and so then there's some citations first in and the ramstead 2018 paper and par and first in in 2019 and so even i looked back through these papers and looked for dynamic equilibrium and it's not there in the references that i was able to find and so i was curious what the authors meant by dynamic equilibrium in a living system i mean as

far as i know um like maybe they meant dynamic coupling between the organism organism and the niche or maybe they mean the non-equilibrium steady state of homeostasis which is not a dynamic equilibrium it's it's a non-equilibrium so i just found this this phrasing kind of curious um and i'm assuming just based on context that they're referring to homeostasis but homeostasis includes plenty of irreversible reactions irreversible chemical reactions and so that's kind of why this was a little bit it makes sense like if we think about instead of the water ionizing and the H^+ plus and the OH^- minus coming back together and the overall values on both sides are staying the same so let's think in the biological case like moment to moment our our mass is staying the same let's just say or some people might maintain the same mass for 30 years but underlying that stability is actually many many you know synthesis of fatty acids and the breakdown of fatty acids and so it's like okay i get it that they're both going both sides at the same rate so there's a stable point but then also it's constituting its own irreversible thermodynamic reactions sometimes and so it's a good question do fristen at all use the term dynamic and are we talking about dynamic couplings which might be at steady states or are we talking about the non-equilibrium steady state of okay brings us to homeostasis sure so homeostasis is the steady state internal internal physical and chemical conditions that are maintained by living systems and there's a lot of variables that are can be included in here uh mass i think actually changes at night like the the carbon that you exhale like you actually weigh less in the morning from like sleeping on my face and like emitting carbon i heard um i don't know if that's really true but i guess it depends on what scale you're measuring um but but i think that ph has to be the same or within like a pretty defined range um osmolarity sodium potassium and calcium ions blood sugar body temperature is a huge one so and and we act we do things throughout the day like eat and drink and use the restroom and put on jackets to maintain to help with our you know homeostatic conditions um internally well unless we're eating at night we're losing water and some other molecules so definitely we're lighter in the morning and then one discussion that we've had in a few different streams is when do we think of homeostasis as reducing uncertainty from a set point like sort of a desired temperature and then we want to just reduce our discrepancy versus when do we see allostasis or almost like anticipatory processes by which biological systems actually take excursions qualified excursions from set points and how do we think about that within the broader history of cybernetics and it's been said that um cybernetics had some challenges with dealing with post-homeostasis so it's easy enough to see how a cybernetic system seeks a set point but then once you start thinking what is the process by which that set point is itself moved around all of a

sudden that challenged a lot of historical cybernetics frameworks so that's definitely what we're exploring with active let's take another dip into a physics or a mathematical idea of a face base i'll read the yellow parts or you want to go for it i'll read the yellow parts and then you'd give any description so in a phase space every degree of freedom or parameter of the system is represented as an axis of a multi-dimensional space so like every point on this spiral has two variables that describe it it's like its value on the x and value on the y and then the phase space trajectory represents the set of states compatible with starting from one particular initial condition located in the full phase space that represents the set of states compatible with starting from any initial condition so just a quick comment or a thought on that so even just in the maintenance of homeostasis like where your you know osmolarity is the same and your blood glucose is the same or just intracellular glucose or atp levels or you know if you just want to look at at just maintaining homeostasis not even including like the potential possible actions that i could take in like moving in a three-dimensional environment or anything like that but but it it becomes very quickly an expansive multi-dimensional space like something that we're not even like here we have like a 3d ish visualization but you know in all these dimensions it becomes very um very quickly computationally intractable because of the dimensionality that of options that we're presented with and just even to maintain a homeostatic set point is is intense yes base spaces are just kind of the descriptive it's literally just the space that something can happen in and then there are trajectories through face space well there's points which just are static values like three comma four and then there are trajectories which are like linkages amongst sets of points and then also out there in phase space are attractors so in the field of dynamical systems in a tractor is a set of states in the phase space towards which a system tends to evolve for a wide variety of starting conditions of the system system values that close enough to the attractor values remain close even if slightly disturbed so a kind of mental model there is like a rubber sheet and someone pulls the middle down a little bit that is now an attractor because things are from near or far are attracted towards it and then if they're at the bottom of that well they're going to resist being moved away from it and so that's sort of a simple ball rolling into a bowl attractor state and there's simple equations that govern that type of movement but in higher dimensional phase spaces there are different geometries of attractors and then also for biological systems it is a challenge to model using attractors but that's what this paper is really about can we use that model of an attractor to talk about biological systems and how they regulate okay so yeah and i i just pulled this image out actually i think it was brought up in the t in the paper and i don't

remember if the original author who outlined these four types of attractor if that was kaufman i think i feel like it was kaufman but it was cited in this paper as well about the four different kinds of attractors and so that's what's shown here is the strange attractor the point detractor the cycle and the taurus um so just in case you were wondering what those kind of attractors looks like it's it's a drawn to points in this kind of way it is also on another phase space of attractors with its own x and y axis so this is a quote from the paper then blue any thoughts would be cool the ergotic density is an invariant probability measure that can be interpreted as the probability of finding the target system m in any state x when observed at a random point in time first in 2012. the assumption of ergodicity is important to get the fep off the ground since it ensures that organisms can be modeled as having invariant characteristics over time so just to kind of relate ergodicity to the phase space so the concept of ergodicity is in a potential phase space that all of the potential points in that space are visited at one point or another like that is ergonomicity and then the probability that you can find a uh the system in that space or it at that point in space corresponds to how much time the system spends at that point in that space if that makes sense it's kind of a circular loop but this is just one slide we're gonna return to ergodicity later on but this is um a topic that blue and i like learning about however if you know more about this it would be great to have your perspective as well as if you're also just learning it and have questions to bring because there's a lot you could learn about ergodicity and the ergotic hierarchy so take a dive and then let us know how the water is also getting fep off the ground i'm thinking does fep start on the ground does it start underground does it start is it a snowflake with wings has it always been that way right exactly it's like the water cycle okay so on to the fep which is of course the target the system of interest of this paper and in active livestream 14 and before then after then we've had a lot of discussions what is the fvp how does it relate to active inference and it seems like we'll be working on those questions for a long long time but a few pieces that we've pulled out as kind of being good um sort of cues to bring up alongside free energy principle and just see them as like cousins or in a network together multi-scale systems ensembles like the collective behavior of multiple particles or other system subunits the nexus of information and thermodynamics which is really like information thermo like heat and dynamics dynamical systems often thermo and dynamics aren't really separated but we're getting at that delineation action policy selection as variational bayesian inference so using information dynamics to study how policies are selected they're sort of fep as it is today and then there's where could we go and develop with this framework which is what we'll be exploring today any general comments before we get into the paper uh

only that you know you we haven't thermodynamics has classically been one word so like i'm starting to just use info dynamics as one word just to be like non-discriminatory you know or just dynamics it's like it's all information it's all thermo so yes exactly okay here we go to the paper this is the first paragraph of the paper and it's very informative life scientists use the term equilibrium in various ways and so we're going to look at three different definitions of and this is one reason why it's important to have ontologies and have ways of using terms that we can if not distinguish at least be clear about in their use so sometimes life scientists use equilibrium to refer to an inner state of death where the flow of matter and energy through a biological system stops and the system reaches a life less state of thermodynamic equilibrium that's sort of the ultimate equilibrium then there's more often they use it to mean homeostasis which is the ability of keeping some variable in a system constant or within a specific range of values going back to these old citations for where the term actually arises from and so that's sort of a living system so there's parts of it that are consuming energy hence not at the ultimate equilibrium but there's also other elements of it at least over certain time scales that are invariant some other time the term equilibrium is associated with the concept of robustness which refers to the capacity of systems to dynamically preserve their characteristic structural and functional stability amid perturbations due to environmental change internal noise or genetic variation so it's sort of like a low bar a medium bar and a high bar of equilibrium ranging from just yeah the molecules are not doing any motion and there's a carcass on the ground two the molecules are churning and there's something invariant about the living system going up into even the idea of robustness which is that the system is doing something to maintain its sort of equilibrium like its balance amidst noise which is maybe seen as a type of homeostasis but not exactly the same so any thoughts on why um these distinctions are matter are mattering or why start the paper this way you know i'm curious daniel like i i mean we've been in the life sciences for 20 some years more maybe um and i've only used equilibrium in terms of life sciences in the first way to like as death um so i'm just curious and perhaps this is the the inner chemist in me but but are you familiar with all of these uses of equilibrium and the life scientists in the life sciences also i think i've seen the second one people might not call it equilibrium but like a buffer equilibrium well the carbonate and the CO_2 in your blood and it's maintaining a buffered a pH equilibrium so that's not death it's a living system that's doing it but it's referring to some chemical process that's being kept at a dynamic equilibrium within the scaffold that a living biological system is providing and um i'd also be curious i don't think we dove into it this kitano 2004 like homeostasis to keep something within a range of values it

sounds a lot like robustness so how are the how are the second and the third definitions different because like anti-fragility is something different where perturbations actually strengthen the system or cause it to develop for example but robustness is a little bit more like homeostasis open to anyone's thoughts we'll see now on to fristen at all over the past 20 years is giving us some historical context since the new millennium theoretical neuroscientist carl fristen and collaborators have developed an account of the conditions of possibility of a certain kind of dynamic equilibrium between a biological system and its environment the core of this account is a free energy principle according to which all biological systems actively maintain a dynamic equilibrium with their environment by minimizing their free energy through inference in action which enables them to avoid a rapid decay into an inert state of thermodynamic equilibrium so here's a few of the citations 2012 13 rams said at all 2018 par and first in 2019 so here are the papes if people want to dive into the references those are always really good places to start but whether you take the authors at their uh manifold dimensionally reduced word or you go into these original papers or you know you send a tweet to an author or you come in a live stream and talk to them what are the key claims of the fep from a rhetoric perspective what does the fep assume and what does it claim which papers are key to fep not what was important in 2005 from 2002 but today what is important so how do we respect that different citations from the past might have a partial picture or use a different formalism and then which papers are contradictory even if only temporary so somebody might make a claim in a 2010 paper you can't explain situation x and then like you know two weeks later first in it all write the response some people have uh they bought the rumor but they haven't heard the news so how do you keep a coherent narrative of the literature as there are errors contradictions disc concordances all kinds of things that just make it less easy than it could be to understand what is happening with fep its proponents and its detractors um so just a thought on this and something that i was thinking about as i was reading the paper and a lot of this is footnoted throughout the paper but the fbp like has evolved over the last 20 years so it didn't start off like in the state that it is now it didn't start off like a completely elaborated um theory and so it would be really cool to develop a timeline of the fep like first concept you know 2006 or whatever and then at what point in time did like the markov blanket come into play and then when did active inference separate from the fep and and to kind of get like a good idea of this situation as it's gone through it's several iterations now great idea blue i hope that in actin flab and other groups we could collaborate to really make that happen because to have the timeline of developments and cross-link that with

the core terms which we'll talk about later today and then also just who was involved and what were the contributions it would just make it tremendously easier instead of just going back to square one sort of a scholar search just when we're curious about hey has there been something written about free energy and x okay so that was sort of the neutral context or at least not so barbarous here's where the authors uh bring in the the gap the free energy differential in the literature and their contribution what actions are they going to take to reduce the differential lot of attention no citations from who a lot relative to what it's not i'm not saying it's incorrect in the paper just it would be awesome to learn what does it look like to have a lot of attention what does a google trends or a book engram viewer search results are we on the up or are we going down from some period of time so what attention what regime of attention has the fep received and from whom the foundations of fep in statistical physics and dynamical systems theory have not been probed yet that's a claim maybe some people will think that they have so we'll look forward to learning about that because two reasons one the theoretical adequacy of the fep and two its practical utility for the study of biological properties depend on the validity of the foundations so it's like we want the house to be beautiful and useful and if the foundations are not strong it will not be beautiful and or useful so this work of the paper as well as all the work around it is related to that gap so a few big questions like what does it mean for a principal fep we can of course have some subtlety with like what's a principle a framework a theory a hypothesis so let's just kind of roll with it together what does it mean for a scientific principle to have foundations in statistical physics what does it mean for it to be grounded in dynamical systems why does that matter why does it matter how it's grounded or who thinks it's grounded in what for the fep and in general and then always trying to connect the generalities with the minute particulars what does this paper actually do and then what work is still needed because it's easy to get caught up in some of these big sweeping multi-decade many person research trajectories and it's helpful sometimes to just separate out what is the paper doing and then what is the bigger picture like the grand synthesis that has to be done between fep and some other field anything here here's from a footnote and it's um to forestall any confusion okay great all my confusions are about to be forestalled i love footnotes like that our focus is on free energy theorists not principalists by the way theorists assumptions that all biological systems at any scale can be modeled within the framework of ergodic random dynamical systems and that homeostasis and or robustness can be defined and studied in terms of dynamical attractors so i don't know if that's a assumption all fep colleagues i don't know if it's a um rather than an assumption but rather a

consequence like there are assumptions that people use but then actually they have come to this claim as a consequence not as an assumption or an ungrounded axiom and while our discussion should suggest that concepts and mathematical representations from statistical physics and dynamical systems theory are sometimes abused in biology it should not suggest that those concepts and representations have no use so it's saying in fact you can have good explanatory predictive models that involve dynamical systems so it's not saying math and biology is hopeless and then also this 2004 paper i thought who do you cite for saying that you know math gets abused in biology so this is robert may a famous biologist in 2004 and this was just very interesting like understanding the non-linear dynamics and this was in the case of a virus and health situation just extracting some of the key memes it's like we're going to have to learn across systems we're not just going to be able to have one default system we need to think about multi-scale for applying and educating in transdisciplinary contexts like having a understanding of viral dynamics is going to require population and ecological understanding as well as protein i venture to predict that the corresponding immunology text will indeed look different in 20 or even 10 years time okay well we're right between 10 and 20 and maybe immunologists are starting to go oh yeah you can't just talk about what's in the what's the white blood cell touching the pathogen there's a bigger picture there's a society and then what are those abuses and they're not always easy to recognize especially because few have the expertise or the confidence or the personal narrative that they're able to ask questions across disciplinary contexts so it's easy sometimes for something to be transdisciplinary and then like the biologists don't question the physics and the physics don't question the biology and all of a sudden something is not really getting questioned so it's totally a great point and this is a really important kind of fallacy essentially where a mathematical model is constructed with an excruciating abundance of detail in some aspects while other facets of the problem are misty or vital parameters uncertain within a massive range and this happens just so many places it's unreal we made a mathematical model of this oh well you know of course we didn't consider other factor abc so it's not that everything has to be a theory of everything but we have to recognize that extremely partial mathematical models in biological systems may be less helpful than people claim them to be and it might be better if people pooled their efforts and worked towards better models any thoughts on that just uh you know i mean i think i'm definitely guilty of uh like i remember back to uh mike levin's live stream when when we were abusing physics so guilty as charged sometimes um but but it's it is important to you know have an open dialogue and to ask questions like if there's a concept or something that you don't

perhaps understand maybe like you feel like you're a beginner i mean i've definitely felt this way like maybe i'm just being dumb but you know and then there's actually a very valid question at the end of that sentence so if you don't understand something be sure to ask until you do i think yep imagine just all the times somebody has said oh fep does that apply to living systems and so yeah it's a done deal well you know it's not a done deal so the conversation is still happening here we get to section two introduce introducing the fep the free energy principle fep says that unless a biological system minimizes its surprise it will rapidly die now what do those citations say do they talk about system death or do they talk about system persistence more specifically fep presupposes a view of biological systems as essentially persists different authors here and for foregrounds the conditions of possibility for the persistence of so that's their top-level take on fep and then to connect a few of the keywords in howie 2020 refers to a system's periodic phase attractive dynamics so attractors in the phase space that have periodicity in the state space to define what it is for a biological system to exist howie writes biological in existence is marked by a tendency to disperse throughout all possible states in state space e.g a system ceases to exist as it decomposes decays dissolves dissipates or alliteratively dies in contrast to exist is to revisit the same states or their close neighborhoods so this is kind of bringing together what is life schrodinger's question ram said it all 2018 with dissipative systems and systems that resist dissipation systems that have a dynamic equilibrium what is the phase space we're talking about these are the big questions we're getting at and how does one disperse oneself exactly and how i think uh you heard that from somebody earlier today somebody was getting dispersed during a practice talk it happens and where's surprise come into play not just the psychological like wow i was surprised it was a surprise birthday party but how do we go from attractors and dynamical systems to this statistical bayesian concept of surprise and attracting sets and lower dimensional manifolds in attracting sets make that make that quote big one more time for me okay so something i i thought was um just unique or or maybe remarkable was it says free energy theorists relate the notion of entropy to the informatic information theoretic quantity of surprise but i think that this is done before the free energy principle arrives like so in information theory that's claudes shannon makes the relationship between information and entropy and surprise so i don't think that that's a something that the fep started doing well two themes we'll return to today and again and again which is like first this shouldn't be a paid versus pape battle this isn't a battle royale for whose hot take will reign supreme we should be collaborating on bigger research agendas and so clarifying the contributions that different

people have made will help demystify and make active inference and fep more accessible because which claims or which contributions were made like 100 years ago not that that makes it more valid but weren't made recently but sometimes when we only see recent citations it can be easy to get swept up but actually the authors of this paper do a good job of going back to some original citations but these are big eternal questions here's a fun paragraph in the paper um because it's sort of like uh i i reformatted it slightly because it it shows that we're in synchronous and asynchronous discussion with Fristen at all so here is like you know him thinking slash saying first insane by finding this is Fristen's claim pseudo first in claim not actually his quote by finding a suitable attractor in the dynamic model life scientists would be able to gain understandings of the workings of dot basically like this is the dream of active inference that if we had attractor phase base all these terms coming together we could just make it work for diverse biological the authors of the paper write to better understand what this means let's examine how such idealized models are developed and why free energy theorists opt to build their model starting from concepts in physics so this is like the dream of active inference and physical groundings for and then the authors do the right thing which is to um to steal person the argument to investigate what its actual strengths are and why somebody would want to make that argument or why this is something that we're motivated to pursue instead of just immediately critiquing it as like not an important thing to want okay all right here is in the red quote from the paper so first in at all 2006 way long ago way way long ago asked us ask us to imagine a snowflake endowed with wings this winged snowflake could move use its wings as solar reflectors or fans and exchange energy with the environment for a much longer period than we would expect of an inanimate piece of ice to keep going under identical circumstance unlike familiar snowflakes the winged snowflake can choose actions which are the means of by which different outcomes are brought about in different states the environment its wings allow the snowflake to bring about outcomes such as lowering its core temperature in response to an increase in air temperature so let's go to first in 2006 and this is figure one so we're imagining they're sort of on the left is like a traditional snowflake model they're sort of it doesn't have any action affordances so it's just being tossed around in the soup and it freezes when it's cold and then when the temperature gets below this critical phase boundary it melts and the system fails to persist so it's like you know checkpoint are we good with systems persisting versus dissipating this is kind of our simple physical example a piece of ice freezing in the sky and melting all right now let's imagine if the ice froze in a way where it had wings slightly different colored wings i wondered what the symbolism was here but it

has these two different wings and that allows it to engage in action on the environment so for example if it detects or predicts or acts as if it's predicting that it's going to head towards a warmer area that's like you know lower down or something all of a sudden it can flap a few times and get into a colder area so here this snowflake just falls but this one is able to flap and stay cold and so through action it's able to persist longer it's precluding a phase transition from solid to liquid through this active exchange with the environment so that's like the winged snowflake model that's kind of a nice thought experiment from yonder to help us understand like the steps from mere physical system to systems that start to engage in actions on the environment especially those that keep persistence of the system here's figure 3 from that same Fristen 2006 paper and this is really an awesome figure because it shows some concordances between actinf even though it it's not even really used in those terms so that's why we need the timeline like is active even mentioned in this paper in it might not be actually but we can see there's this perception action cycle and this is related to a few other kinds of action perception cycles um footnote paper incoming we can think of this as like a bowl and then the ball is rolling to the bottom of the bowl it's a image we return to again and again now the y-axis is the free energy and so here's the minimum possible free energy and it's like the system starts here there's a change in the environment that causes a mismatch that mismatch can be reduced through two different types of updates of a model one is an action and the other is a perception so which order happens details details details but the idea is like at each loop there's an environmental change which is then renormalized or returned to something through perception and action which relates to an act of inference how we say that the two ways that free energy can be minimized are through updates of the generative model like learning or parameter updating or through action how do we reduce surprise inference and action why reduce surprise because if we have an optimistic generative model we don't want to be surprised about our model where we're living so we act and we infer so that we persist any comments here but it's just a cool like input could be the state of chemoreceptors you know already we're going from these sort of physics formalisms and dynamical systems formalisms towards just oh yeah you know drop it it could be like chemical biological was it justified in 2006 well looking back on it not as much as it is justified today and what will they say in 2033 looking back on this so all work in progress so this is from page six of the paper won't uh read the full quote because we've kind of gotten at this trade-off between utility and accuracy and generalizability but the big questions here are what systems is the fep modeling or describing where has it been used where could it be used where shouldn't it be used or can't

it be used how good is the fep for modeling or describing these different systems how good could it be and what would it take to get it there and then also an important question for those of you who are fep or active fans and you have a challenging conversation take a step back and ask your colleague what models do you prefer with similar or different explanatory scope just what else is out there because it really helps ground the discussion in what that person's understanding or preferences are and a lot of times people will come up blank they'll say i'm critiquing your unified model of perception cognition action okay great critique away that's what i'm doing too what do you prefer up nothing so it helps us stay positive with the contributions that we're making to this field while also recognizing the discrepancy and that combination of recognizing discrepancy and taking action is active inference you know this is funny this is like a good contrast to the sims paper that we just did and i feel like at the end of the last live stream we were just talking about like thermodynamics and information theory and the relationship between those two things um but i always want to take the fvp and use it for like non-biological systems i think i like veer toward pan psychism more than most people and like i don't like that i have a hard time with the line between mirror active inference and adaptive active inference or like the evolutionary late evolutionary cover slate whatever uh like the you know brain is the ultimate um source of cognition and capable organ capable of cognition and so like would the fvp apply to a set of molecules in a beaker like it doesn't have preferences did molecules prefer to be in an equilibrium state or how does that work out like i want to take the fvp and use it in systems that maybe it's not designed for i don't know i tend to want to go the other way makes sense here's that adaptive versus mirror active inference slide from 30. good segue perfect segway and um so here's a question from fep landia what makes biological cognition special or different we're not going to go too much into it here we talked about it in live stream number 30 a ton and it was just super interesting to what extent does it matter that it can engage in counterfactuals does the temporal depth matter but if we're on the outside how do we know what the temporal depth is is it the temporal depth it acts as if it's cognizing where does energy dissipation come into play so these are big questions what is cognition and is there just one kind of cognition and biological systems do a special combination of it but it's different in kind of decimal point but not in the top level like are they or are they really different in type or is it such a great quantitative difference that it's different in type big questions here more from the paper active inference models are phase space representations of biological systems so that's like a full stop moment this isn't what biological systems are it's a phase space representation of systems here we're talking about biological

systems as forming expectations over observable external states and inferring policies state action mappings that minimize the expected free energy of those states under a generative model in some predefined markov decision process by minimizing expected free energy the modeled system would attain a non-equilibrium steady state and so it would maintain in some sense a low entropy probability distribution over its states so what is active inference what topics is it related to what is it based upon we look at that ball in the bowl and it's easy to say well the way that the imperative that's pushing the system forward is that the ball wants to be as low down as possible and it's going to be minimizing its potential energy that way another story about the bowl is the ball is going to end up or act as if it's going to end up with the tightest bound through time on how far left and right it goes now that also takes you right to the bottom of the bowl but you didn't have to mention that you wanted to go down it was just that the desire to minimize the uncertainty led to the bottom of the bowl in that structure how do we think about some of these different ways and what are the physics metaphors an assumption from the paper an assumption made by the active inference models developed by free energy theorists is that target biological systems can be represented as random dynamical systems what does this assumption mean exactly what does it mean and what are the assumptions of active what are the assumptions of other frameworks does active have the same assumptions as other frameworks does it have fewer or different does it have more assumptions are there assumptions of active there are consequences of other ideas or vice versa so these are just awesome questions that the authors are raising and that their work is like a jumping off point into good good i so so one question that i always have is what if the so active inference and the free energy principle presumably have the same assumptions but but do they have to um and so that's always something i wonder how has active inference developed this is a fun paragraph active inference modeling evolves rapidly eats shoots leaves as maxwell ramstad dr maxwell ramsay helpfully reminded us in conversation up to roughly 2012 the dynamics of internal and active states inactive inference models were determined through gradient descent on variational free so let's call that phase one before 2012. we need a real timeline for this this is just informal but this is why we need the timeline figure out if we're truly on the best timeline from around 2012 until recently like 10 minutes ago or 2019 active inference models have been equipped with algorithms for policy selection which evaluate the average free energy expected under each policy so expected free energy not just variational free in their latest work the active inference community have endowed their models with a recursive expected free energy functional which enables the models to represent target

systems higher order counterfactual beliefs which are beliefs a system has about the beliefs it would have as a consequence of action so that's the deep parametric active inference that's sophisticated active inference that's what we spoke about with a mental action paper sandved smith at all okay so those are three waves that are important to distinguish in the development of active inference models because these modeling advances all seem to make assumptions about the ergodicity of biological systems and the biological meaning of dynamical attractors the points we make in what follows may help explain why hashtag all active inference models thus far tend to trade off realism for maximal generality so they're respecting the developments on the framework and encompassing it summarizing it maybe accurately maybe not and then saying because they all make some common assumptions they're all victim to this trade-off of realism for generality or is it utility for generality which one was it again realism utility what are the trade-offs so what are the waves of active inference how does aerogenicity come into play in these different waves i know there's a lot on there but this is really a important piece to understand that there's been recent developments yeah it's still in progress here um this is some technical details that would be awesome to just learn more from the authors or from anyone else who sees this and gets excited by it this is about some of the physics formalizations of noise and dynamics and phase spaces and just a term that i was curious about is like a co-cycle so it's kind of taking that idea of returning to an attracting state and maybe connecting it to some other ideas like a co-cycle but these are just things to um learn and discuss about this is sort of the classic bayesian network markov blanket is there a markov blanket that picks out the mind is the network of autopoietic processes and biological systems identical to the markov blanket of the system what is the metaphysical status of markov blankets and then the authors write these debates in the metaphysics of the fep can be left on the side since we're not interested in metaphysics here i thought we were having a discussion that was meta to physics where is your metaphysics now okay anything else there action in the loop given this setup free energy theorists are interested in finding the active states of the systems that are confined to a bounded subset of states and remain there indefinitely so we want to have our internal states be at a certain level like thinking about our temperature being at a certain value or anything else and now we want to talk about the control theory question which is what active states should the system engage in conditioned on internal and sensory states such that the internal states do remain bounded within a subset of states more uh technical details if one assumes that the dynamics of the target organism is dynamics is ergotic if one assumes that the dynamics of the target organism is ergotic it's grammatically incorrect if one assumes that the

dynamics of the target organism are ergodic that's why you can't read it that's that's why the compiler failed then that's it random dynamical attractors can be associated with the ergodic density a p -distribution of x given m which is proportional to the amount of time each state is occupied by the organism so here's where we return to their definition of the ergodic density as the invariant probability measure μ there's a few other terms that I know some listeners know about whether they'll help us with it's another but what does it mean to talk about the Lebesgue measure of the attractor how is that connected to dynamical systems and entropy and whatever this integral is calculating in the limit of time approaching infinity but this is kind of this thing that we see in space and in time one over space like what's the tiniest infinitesimal space for all space or what's the tiniest tiniest moment for all time how does that connect to action control theory more more on that later but here even in this μ even in this slide it raises the so here it says such active internal states in an organism and so maybe it's the way that the diagrams are usually drawn for active inference but I don't ever think of active states as internal states like there's active states sensory states internal states external states and I always think about the sensory and the active states as the on the boundary between internal and external absolutely there μ are internal external sense and action states so what are the active internal states there's a few ways of thinking about it the particular states are the sense action the blanket states and the internal states so maybe that's what's being referred to or we can well when we're talking about mental action I guess then the active states are internal between the past and the future but still eternal right like in mental action right yes with this principle with this definition of active states in hand so I believe that we're now working with the Fristen ontology of the action being not the internal states but who knows first in 2012 finally formulates the principle of least action as follows the principle of least action the internal states of an active system so that wouldn't be the action states that would be the internal states of a system with on the interface having action they minimize surprisal I such that the variation δS of action S with respect to its internal states r of t existing in r vanishes and then classic future Fristen look at that the 2100s are going to be just so good the FEP would suffice to derive the principle of least action and is defined by Fristen in the past and the future as follows the free energy principle technical technical technical if the internal states minimize free energy then the system conforms to the principle of least action and is an active system this will be good to unpack and learn more from the authors as well as from those who studied these areas because it's sort of something we all ask is like wait least action but then they're engaging in action couldn't they just do less action wouldn't that be more of a fulfillment of the

least action principle but what is it actually it's like a stationary principle it's an invariant so there's a little bit more nuance than just staying on the couch and i haven't heard this term active system before i've heard like active inference system or inactive system but this active system is new to me and indeed Friston was a very forward thinking guy uh he is a very forward-thinking guy all the way into the 2100s when expressed in these terms so in the terms of physics the FEP provides us with an account of biological persistence as a generic type of stability of random dynamical systems so then they talk about first in 2012 and 13 some simulations and then here's what the author's challenge is and say let's then put into better focus some of the difficulties involved in this challenge and ask is there any reason to believe that all biological systems at any scale can be modeled as random dynamical systems and how does this maximal generality bear on the degree of realism of the models so we had a few previous discussions ago we had um we had realism we had utility we had instrumentalism we had all these different axes and how are we going to manage that trade-off because sometimes the most quote realistic description of a system isn't the most useful or it's not the most general so there's a lot of trade-offs happening how does model realism relate to generalizability and how does that relate to the utility of the model as well as other attributes like its simplicity that's like the relationship between the map and the territory if you wanted the territory then you wouldn't need a map yeah and then almost like the compass um or a piece of paper that just you know says like you are here it's like it's so general you can always break down and that map is going to be at least not telling a lie you know this is your left or something like that this is is that useful it generalizes really well but have we lost this whole question of navigation in the desire to find the simplest map so something that that i wonder about like um you know our biological systems random dynamical systems like what's a non-random dynamical system like a dynamical system as opposed to like a static system like statics dynamics and so like what other system could a biological system be like if it's not a random dynamical system what is the alternate of a hypothesis just like what you said about like okay well what is the better model for cognition if not you know active inference then what model is the better model right like so but what about for a biological system if it's not random dynamical system what is the best model for a biological system dynamical through time dynamics random used in a lot of different ways but is that statistical is it stochastic where is determinism fit into play well let's talk about it so here's section four and this is like some of the main body paragraphs and contributions of the paper it's the section modeling biological active states as random dynamical attractors free energy theorists account of

biological systems persistence relies on three main modeling assumptions the existence of markov blankets that imply a partition of states into external and internal the existence of random dynamical in this section we concentrate on assumptions one and three and on the more fundamental challenge of defining phase bases for target biological systems in the life sciences so the three core ideas that they're saying are underlying fep work in biological systems are ergodicity markov blankets and random dynamical attractors are those the main three that's a really important question this paper is going to focus on the first and the third not going as much into the markov blanket discussion because there's been a huge amount of discussion on markov blankets and we've talked about them severally so they're going to focus on ergodicity and random dynamical attractors that's 4.2 and 3 but that's going to be introduced by a discussion on phase spaces in biology so that's what the authors have set out to do um shameless lab plug we're working on an active inference ontology in a participatory way that includes references definitions translations into multiple human languages it will be many things and it will scaffold educational content that will make this entire field more accessible and rigorous so terms like face space um we've talked about we talked about state space on september 13th so every week we're bringing together different perspectives and sharing resources and working towards common understandings in this document to really get at what are the core assumptions of fep or active what do these terms mean what would somebody need to learn if they wanted to understand more so it's really important work and we're definitely going to draw upon the author's contributions here as we work to scaffold an active inference ontology so that we don't get tied up in active state action state systems with action we can be clear about how we're talking about this field and in fact we really have to be anything on our ontology document before we continue back to the paper but this has been a fun project with many contributors so let's talk about phase spaces in we introduced this in the earlier keyword section so the phase space the system is a giant space formed by the relevant degrees of freedom of the target system that's the space of the possible a point in the phase space corresponds to microstate which completely determines the system in terms of the variables and parameters required for analyzing its dynamics and the dynamics are determined by the equations of motion which describe the evolution of points or trajectories in so here's that phase-based kind of model and then the authors are going to suggest that there's at least three facts cold hard facts that make face spaces in biology hard to apply because first biological systems have many degrees of freedom it's not just a pendulum second the symmetries which are invariant preserving transformations underlying the observed

regularities in biological phenomena are more unstable than physics and third the phase space for many real world biological targets is much less stable than that for the kinds of target systems studied in physics so what do you think about that or about phase spaces in biology yeah i mean i think i mentioned the multi-dimensionality in the beginning when we were talking about the phase spaces in biological systems um and i think that it just comes to that with any um with any system right like you can always include too many parameters like you can have the territory or you can have a dramatically reduced map and so finding that that correct balance is hard and if we're looking at um you know say states of a system and trying to predict the next state are all of the factors conditionally independent is there some you know mingling of the variables there and is it okay to just model a couple at a time and what if we don't know if there are some confounding variables in the system too so it's it's a difficult task but it's how do we start to get a grip on it as hard yes and a big question in the background does this matter for applications so yes these are in principle critiques of doing dynamical phase space models of but we know that dynamical systems models of neural systems of hormonal systems ecology these are all really useful and people use them every day so gene regulatory networks don't forget those how could we forget but does something having these sort of fundamental critiques prevent it from being useful not always this is a really nice um points about the open-endedness of biological phase space so this is the work of longo montvale and kaufman 2012 who said that in physics one can pre-state the phase space for target systems based upon stable stable invariances and symmetries so we can like say the ball is going to be rolling around in the box those are the boundaries and here's the invariances historical processes studied in evolutionary and population like all these processes that we know and love in biology involves symmetry breaking which makes the phase spaces structurally unstable ever-changing and unpredictable now unpredictable is a little bit of a tough one because that's a continuum what if you can predict one percent of the variance in great evolutionary transitions is it unpredictable or is it predictable you know is it half full or half empty and then this is from the paper because we cannot pre-state the ever-changing phase base of biological evolution we have no settled relations by which we can write down the equations of motion of the ever new biologically relevant observables and parameters but that we cannot pre-state more we cannot pre-state the adaptive niche as a boundary condition so could not integrate the equations of motion even were we to have them all so it's like there's the ball rolling around on the billiards table but now there's mutations in the genome of the ball so that there's like totally new mechanisms arising in a way that might have a historicity

oh yeah and the table is mutating and it's made up of other organisms so it's easy to see how we get lost in the complexity yet adaptive action exists and useful models exist so how do we reconcile the fundamental open-endedness and creativity and novelty of evolution with our desire to explain some portion of the variance so we can act effectively well and this relates like to a question that i had brought up to you um yesterday daniel i think about like what about discretizing space and so i know that like some of the physics principles rely on infinite time like the ergodicity specifically relies on the possibility of infinite time and this is all stuff that's like kind of beyond me but but what if we have a temporarily restricted moment right like what if we're just interested about now and five years from now or what if we're just interested in now and five years ago right and and so in each moment is this symmetry breaking happening or at what scale and so is there perhaps a chunk of time a small discrete chunk of time from you know a microsecond to 100 years where these types of um inferences can be used and where the equations are applicable and where we can use the model with under the you know specification that this model is not good for the next two million years but it will last you 20. right like so it's a model of the phase space of blood glucose versus ph in a person as measured by the devices we have it's not of blood sugar over a million years because that doesn't even make sense given the system of interest so how do we reconcile pragmatics and utility with these really important philosophical questions so that was four one face spaces word two one of the assumptions of the fep is that all biological systems are ergodic paging ramstead at all at all this is something we really want to unpack and understand like how how does it mean ergodicity or more precisely metric transitivity in random dynamical systems like what is the math and the physics and is that an assumption of the fep what does this mean to assume that biological systems quote are ergotic which is different than saying biological systems for some spatial temporal scales can be modeled using statistics that involve ergotic assumptions so there's a lot of nuance there like whether we're making realist claims about how biological systems are or whether we're using statistics in an instrumental fashion to make useful models of organisms but the big questions are like what kinds of systems are ergodic and how what are the implications of ergotic assumption is it ever not an assumption but it's actually like a consequence of some empirical grounding there's the ergotic hierarchy so ergodicity it's not just like yes or no there's a total continuum with a lot of complex math and then just for everyone how or where does this matter for modeling under the fep or active inference in real world situations and they did call out in the paper um a reference to ramstead and the local ergodicity which is you know for a biological organism i'm like something like

the you know boundaries in which the organism can exist um anyway i think uh maxwell come come explain it to us please but um the authors did mention that so here's a quote from the paper birkhoff 1931 demonstrated that if a dynamical system is ergodic then the infinite time limit exists and coincides with the phase average for almost all initial conditions little x like the realized conditions within the bigger x like the space of the possible this is called the ergodic theorem and that is going to get linked to everything we were just discussing surprisal minimization as ergodicity maintenance and then here the proof of the principle of least action is straightforward and rests on noting that action and the entropy of the ergodic density over external states are related via the ergodic theorem firsten past and the future pnas burkhoff 1931 pretty interesting paper and just it's amazing to think like how in 100 years of mathematics after this but this is kind of that initial formulization of what if the space average and the time average were going to be related to each other so it's a cool topic we've brought up ergodicity before and it would be really helpful for people who know some of the technical details to help work with it this is one um analogous to how they brought up like here's the facts that challenge phase spaces in biology so here's some facts that challenge the application of ergodic theorems to real and there's good reasons the authors say to think that the theorem cannot be realistically applied in the domain of biology so first off it's not enough to show the infinite limit exists it also has to be shown that the convergence rate is plausible we're going to talk about that in a second second point comparing biological to non-biological systems it's harder to estimate the amount of time that they need to kind of asymptote because of the non-uniformities another reason and this is going to be one that i know people will have things to say why ergodicity cannot plausibly be assumed in biology is that the ergodic theorem requires the dynamics of the system to be ergodic which means that eventually almost every point will visit every measurable region in x we've talked about that before like but biological systems never revisit the exact same state if you think about every molecule constituting the phase space but if the phase base is just temperature versus blood sugar yes you pretty much will revisit some points but will you revisit every point are you going to reach a value of negative 5 000 sugar and 900 degrees celsius no we're talking about bounded states that's kind of a nuance and then well and also it's also the arrow of time is is moving forward right it points one direction and so like you know at age 40 my skin will never have the collagen it had at age four like that's never going to happen again right so i mean and there are some things that just decrease as you age so there are yeah let's look at this palacios 2018 paper so this is i guess an earlier paper by the first author so it kind of gives a little context had we but world enough and

time but we don't exclamation point colon justifying the thermodynamic and infinite hyphen time limits in statistical mechanics great title well punctuated uh a bevy of punctuations and this is just a very interesting paper that we skimmed over and i'm sure there's a ton to say it'd be great to speak with the author about it but they argue that this has there's an important consequence for the philosophical literature on infinite limits and that's basically it's not just about like where you are that's where you are in phase space it's about where you're going that's the asymptote in infinite time and it also matters how fast you're going to get there so it's like kind of the asymptote for a pane of glass is like in a puddle on the ground because it's a liquid or it's a glass but that might take so long that over some time scale it can be modeled as if it's not changing for example so that's kind of a cool notion that it's not just about where a system is or where it's being attracted towards but actually the dynamics of the approach matter a lot and that's certainly true in biological um the assumption of ergodicity to give a definition of equilibrium states is a controversial assumption even in the domain of physics especially due to existence of physical systems in equilibrium that have proven to be non-ergodic so i didn't find this palacios 21 citation in the biblio this is the reason why physicists and philosophers have offered alternative approaches to equilibrium that do not rely on ergodicity even philosophers that defend a quasi-ergodic approach to equilibrium recognize that this definition of equilibrium may plausibly apply only to a restricted class of systems such as gases sounds bad but then we go okay the authors say it doesn't mean of course that ergodicity or a weaker notion of local ergodicity may sometimes be a reasonable assumption to make so in other words it might be useful for specifically defined systems anyway however the fact that one cannot extrapolate the assumption of ergodicity to the behavior of all relevant systems in biology cast doubt on the project of formulating a principle that depends on this assumption and is both maximally general and biologically realistic so ideas that we've heard a little bit about but how are we going to finesse realism generality and grounding in physics when biological systems might be grounded and things are not fulfilled so that's 4.2 all right 4-3 random dynamical attractors what makes the application of fep to account for real-world behavior is even more challenging beyond 4.2 and 4.1 because one also has to show the existence of an attractor and also justify why certain attractors but not others plausibly denote homeostatic states in the target there might be multiple attractors and then there's this question of chaotic systems do you want to add anything on these basically it's non-trivial to say that there's an attractor and it's the right one and it results in adaptive behavior there's a lot that goes into it we can learn more we'll hear more questions later and then just to go quickly to

five and six so five discussion in the previous sections we've analyzed three challenges involved in the justification of the free energy principle so one was pre-stated phase spaces for most biological systems two was the lack of the warrant for making ergodicity assumptions in and three is the challenge of identifying homeostasis with attractors in a phase space even if one and two were accomplished based on our analysis one overall conclusion is that because of a fundamental mismatch between its physics assumptions and the properties of biological targets model building grounded in the fep achieves maximal generality yay for minimal biological plausibility but how is that going to work as we raised earlier how does generality versus plausibility get traded off for real systems in the fvp in active and in what other frameworks let's always have another one if people want to or can think of one so that we're not just asking whether it's like as good as it could be because here's a hint it's not as good as it could be we're always going to see ways to improve the theory and the literature if only by organizing other work and making it more accessible what is its relative biological plausibility what is its relative generalizability and what are we relativizing too because as they say like for other idealized scientific representations even if the fep involves simplifying distortions it does not follow that it must be minimally realistic and have limited explanatory or predictive power so they're saying like it's kind of like enemy my enemy is my friend it's like saying just because you idealize and simplify that it doesn't mean that you're gonna get explanation or prediction that's their discussion they fall back to the three key points that they think challenge the usage of physics ideas in biology that are postulated to be underlying the fep phase spaces ergodicity and random and they're concluding that there's a mismatch between the physics assumptions and the properties of biological targets here in the implications and predictions they write if the fep or active inference models it grounds make simplifying distortions distorting assumptions about the phase space of a target system the ergodicity of biological systems and the existence of an attracting set corresponding to homeostatic states for these systems then these idealizations should earn their keep so you know you've been a bad theory but it's not all over you could be a good theory you could earn your keep maybe in the future and then they um write these idealizations should allow life scientists to attractively draw some explanation and prediction about relevant biological observables in real world systems and assess those predictions against measurements relevant to understanding some aspect of homeostatic processes in actual de costa at all 2020. our contention is that the idealizations made by free energy theorists do not play these practical and epistemic roles and so this is the de costa at all 2020 active inference on discrete state spaces a synthesis here are from table

one here's all the applications so it's interesting because like this pragmatic and epistemic role that's kind of the ontology of active inference that's how we talk about policy selection and then it's like okay looking at this list i'm seeing a lot of pragmatic and epistemic value what is the contention that the idealizations themselves do not play pragmatic or epistemically valuable roles i think it'd be good to get a little clarification because table one has a lot of and since then even there's a lot of interesting areas what do you think about that blue yeah it's interesting like the word choice of the author here like the idealizations made by the free energy theorists theorists so what are these idealizations like the reliance on ergodicity and random dynamical systems theory like are these idealizations or is it actually using the fep and active inference for practical applications that don't play the practical and epistemic roles so i just i'm curious as to that word choice and steven in the chat wrote the use of approximation science around ergodicity could give the potential to have sufficient revisiting of states to make meaningfulness plausible how can we measure when sufficient approximation to ergodicity exists in order to make variational free energy inferences so good question we can ask and as we kind of explored revisitation is one component of ergodicity but also there's this element of re of revisiting every point and of exploring the whole phase space so as we kind of addressed what is the reality of ergodicity and applying those models to biological systems and then yeah can we use approximations and approximation science to make it work even if in the infinite perfect case it doesn't exactly seem like it would like the t-test assumes that there's an equal variance between the two groups unless you have a t-test within equal variances yet no two groups in reality are going to have the exact same variance yet the t-test works so how can we take that same approach and think about the systems that we're working with here here's their whole conclusion in a churning mail storm they're arguing that fep theorists have pursued maximal generalization by relying on physics concepts however that means that they've sacrificed biological realism and the danger of sacrificing biological realism is that you risk minimizing explanatory and predictive of those accounts for biological systems which are homeostatic and manifesting far from equilibrium in their persistence and their last paragraph the fep can perhaps be better understood as a maximally general definition of any system that persists but this definition does not seem to provide us with any new insight into to the extent free energy theorists treat all biological systems at any scale as pre-specific pre-specified generic objects with fixed currently unknown equations of motion their account risks missing all features that make biological systems interesting kinds of thermodynamic systems so it's like you know a city where there's all kinds of intersections and then you have a

grid map and so we distort the system in a way where all of a sudden we thought we were generalizing but we ended up losing touch with the system and therefore explanatory and predictive any thoughts on the conclusion um no i mean i think that that you know they have made their point pretty clearly so um i think the fep also has a lot of room to grow being that it's you know relatively not not quite 20 yet so i think that um you know hopefully some pragmatic and epistemic value will be put forth by um the generalizations made by the fep in some future model perhaps that's almost like a deep time sociological value even a provocative theory which has limited epistemic or pragmatic value not saying the fpp is could play a role in a broader social system that still gets the job done so that's kind of a cool notion what questions for 31.1 did you put up so i think i raised them all already so like what is the dynamic equilibrium of living systems as the authors state here and as i stated in the beginning keywords and it says we should pay closer attention to the fundamental differences between physics and biology like what are the differences between physics and biology like in the paper they made they went to a great length to explain like why biological systems weren't ergotic but then also like they made the point that physics they have they have a really hard time like proving ergonomy exists at all um in any physical system so so what are what are these fundamental differences between physics and biology that that are so critical like that it seems that you know the fep kind of misses the point um and then they refer often in the paper to active states as as internal states and i i just wonder what like i mean i've really thought about active states as like the interface between internal and external and part of the blanket and so i just wonder why the active states are included in internal states and their you know description of the model so so just some some technical things that i have questions about and um you know hopefully the authors will be here to answer them cool stephen wrote we may never know the whole phase space as the phase space may be variable yes that's what we talked about like the phase base of temperature does that include impossible temperatures that the system cannot persist at and then the fact that the phase space can change is the open-endedness of the um the longo i think 2012 open-endedness of the phase space in biological systems so for sure that's why we can read the papes and then have specific citations and ontologies in mind so we're not kind of cross angling because those are great points and they were the points made in the paper and then stephen asks is biological realism based upon folk biological assumptions so i think this sort of like folk hyphenated everything has come into the forefront with the discussion with matt sims so yeah what is biologically real you know what models are biologically realistic isn't that just like the question of what is real so yeah what is

even real at all exactly yeah realism sounds good until you so you ask too many questions
what a fun discussion so we ask all these questions like what does a good understanding
enable what are the unique predictions and implications of this paper what are the next steps
for fep and active what are the goals of this research and then what are we still curious about it
was a fun discussion so thanks a ton blue for doing it and for working a ton on the slides we'll
look forward to 31.1 and beyond awesome thanks bye bye